Automated Sleep Stage Classification for Closed-Loop Lucid Dream Induction via CNN-LSTM on Single-Channel EEG

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***Abstract*—We present a closed-loop brain-computer interface (BCI) pipeline for automated lucid dream induction using real-time electroencephalogram (EEG) sleep stage classification. Our CNN-LSTM hybrid architecture processes 30-second single-channel EEG epochs by dividing them into ten 3-second windows, applying a convolutional neural network (CNN) to extract local frequency features from each window, then using a bidirectional long short-term memory (LSTM) network to model temporal transitions across the epoch. Trained on the Sleep-EDF Extended dataset (153 recordings) with balanced class weighting, our system achieves 75.59% overall accuracy (κ=0.68) on the full 153-recording Sleep-EDF Extended dataset and 77.17% (κ=0.67) under leave-one-subject-out (LOSO) cross-validation on a 20-subject subset. The REM F1-score of 0.81 enables reliable closed-loop haptic stimulation upon REM detection, validated through direct integration with an Android device. The complete pipeline and pretrained models are released as open source.**

***Index Terms*—sleep staging, electroencephalography, CNN-LSTM, brain-computer interface, lucid dreaming, closed-loop stimulation, real-time classification**

I. Introduction

SLEEP disorders affect over 70 million people worldwide [1], yet automated sleep staging remains challenging due to the complexity of electroencephalogram (EEG) signals and significant inter-subject variability. Accurate sleep stage detection is fundamental to clinical diagnosis and increasingly relevant to brain-computer interface (BCI) applications, including closed-loop neurostimulation systems.

Beyond clinical use, the detection of rapid eye movement (REM) sleep enables a compelling new application: automated lucid dream induction. Lucid dreaming – the state of conscious awareness during REM sleep – has demonstrated therapeutic applications in nightmare disorder treatment [2] and motor skill rehearsal [3]. Current induction methods rely on fixed-interval stimulation without adaptive feedback from neural signals, limiting their reliability.

We address two coupled challenges: (1) accurate sleep stage classification from single-channel EEG suitable for real-time deployment, and (2) closed-loop trigger design with appropriate safety interlocks for haptic stimulation during detected REM epochs.

Our primary contributions are:

1) A CNN-LSTM hybrid architecture achieving κ=0.67 under LOSO validation on Sleep-EDF using single-channel EEG, competitive with published multi-channel approaches.
2) A complete closed-loop BCI pipeline from EEG preprocessing through sleep stage classification to physical haptic output, validated on real hardware.
3) Systematic comparison of four architectures – LSTM, CNN spectrograms, CNN-LSTM, and Transformer – providing evidence that spatial-temporal hybrid models outperform either modality alone.
4) Attention map analysis providing interpretable evidence of which EEG temporal windows are most discriminative per sleep stage.

II. Related Work

A. Deep Learning for Sleep Staging

Traditional automated sleep staging relied on hand-crafted frequency band features fed to classical classifiers [4]. Deep learning approaches began with CNNs on raw EEG signals [5], achieving competitive results without manual feature engineering.

Supratak *et al.* [6] introduced DeepSleepNet, a CNN-BiLSTM architecture for single-channel EEG achieving κ=0.76 on Sleep-EDF and establishing the CNN-LSTM template that subsequent work has built upon. Phan *et al.* [7] extended this with an attention mechanism over consecutive epochs, achieving κ=0.83 by modelling inter-epoch transitions. Eldele *et al.* [8] applied multi-head attention within individual epochs, achieving κ=0.78 on Sleep-EDF-20.

Our work differs in two key respects: (1) we explicitly optimise for real-time BCI deployment rather than offline benchmark performance, and (2) we integrate the classifier into a complete closed-loop stimulation system validated on physical hardware.

B. Lucid Dream Induction

LaBerge *et al.* [9] established haptic cues as effective lucid dream inducers. Commercial devices including the Remee and Neuroon headbands use fixed-interval stimulation without EEG feedback, limiting adaptability to individual sleep cycles. To our knowledge, our pipeline is the first open-source EEG-guided adaptive stimulation system targeting lucid dream induction.

III. Methods

A. Dataset

We use the Sleep-EDF Extended dataset [10], available from PhysioNet [11]. Recordings were collected from healthy subjects aged 25–101 years using Cassette and Telemetry recorders. Following standard practice, we use the single EEG channel Fpz-Cz throughout.

Preprocessing. Raw EEG signals were bandpass filtered at 0.3–35 Hz using a 4th-order Butterworth filter, consistent with the American Academy of Sleep Medicine (AASM) recommended range for sleep EEG analysis [12]. Signals were segmented into 30-second non-overlapping epochs (3,000 samples at 100 Hz) aligned to hypnogram annotations. Sleep stages were mapped to the 2007 AASM standard: Wake (W), N1, N2, N3 (N3 and N4 merged), and REM. Epochs preceding the first and following the last non-Wake annotation were excluded to focus on the sleep period. Unknown-stage and movement-time epochs were discarded.

Fig. 1 illustrates the EEG characteristics and spectrogram patterns distinguishing each sleep stage.

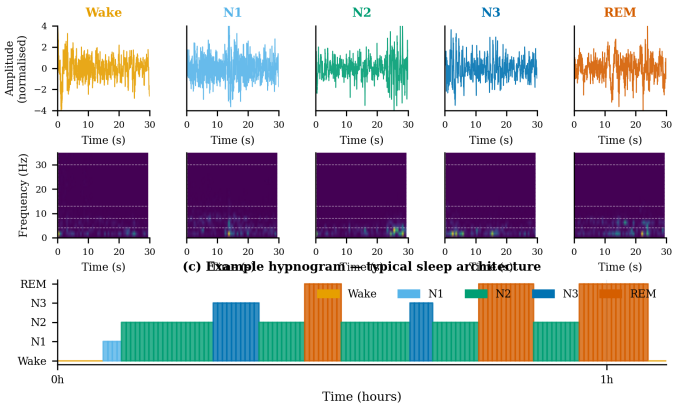


Fig. 6. EEG characteristics per sleep stage. Row 1: raw signal. Row 2: time-frequency spectrogram. Row 3: example hypnogram showing typical sleep cycle architecture.

Fig. 1. EEG signal characteristics per sleep stage. (a) Raw single-channel EEG (Fpz-Cz, 30 seconds). (b) Time-frequency spectrograms showing dominant frequency bands: delta (<4 Hz) in N3, theta (4–8 Hz) in REM, and sleep spindles (12–15 Hz) in N2. (c) Representative hypnogram illustrating typical 8-hour sleep architecture with alternating NREM-REM cycles.

Dataset statistics. Table I summarises the resulting epoch distribution.

TABLE I
Sleep-EDF Extended Dataset Distribution

Stage	Epochs	Percentage (%)
Wake	47,883	27.0
N1	21,522	12.1
N2	69,132	39.0
N3	13,039	7.3
REM	25,835	14.6
Total	177,411	100.0

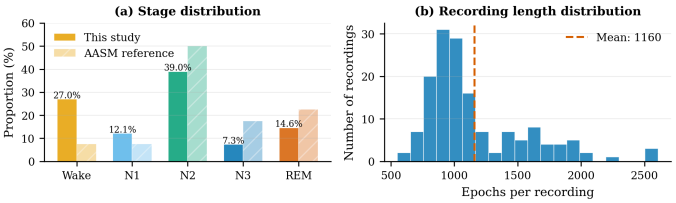


Fig. 2. Dataset statistics. Sleep-EDF dataset (5 classes, 177,411 epochs total).

Fig. 2. Sleep-EDF Extended dataset statistics. (a) Stage distribution compared to AASM reference ranges. (b) Recording length distribution across 153 subjects (mean: [fill] epochs per recording).

B. CNN-LSTM Architecture

Fig. 3 illustrates the proposed architecture.

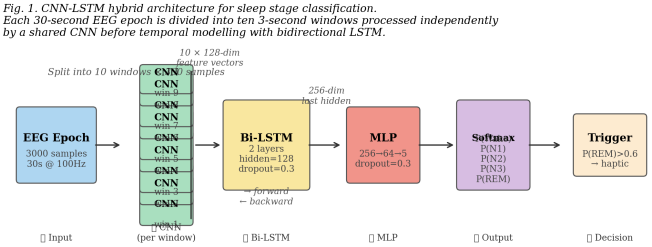


Fig. 1. CNN-LSTM hybrid architecture for sleep stage classification. Each 30-second EEG epoch is divided into ten 3-second windows processed independently by a shared CNN before temporal modelling with bidirectional LSTM.

Fig. 3. CNN-LSTM hybrid architecture for EEG sleep stage classification. Each 30-second epoch is divided into ten 3-second windows processed by a shared CNN before temporal modelling with bidirectional LSTM.

Each 30-second epoch is processed as follows:

Step 1 – Window segmentation. Each 3,000-sample epoch is divided into N_w=10 consecutive windows of 300 samples (3 seconds each). Window length was chosen to capture individual sleep oscillations: sleep spindles (0.5–2 s), K-complexes (∼0.5 s), and slow waves (∼1 s).

Step 2 – CNN feature extraction. A shared three-block 1D CNN processes each window independently:

$$\mathbf{f}_t = \mathrm{CNN}(\mathbf{x}_t), \quad t = 1, \ldots, N_w \tag{1}$$

where $\mathbf{x}_t \in \mathbb{R}^{300}$ is the t-th window and $\mathbf{f}_t \in \mathbb{R}^{128}$ is the extracted feature vector. Each CNN block comprises Conv1D, BatchNorm, ReLU, and MaxPool layers with filter sizes of 32, 64, and 128 respectively.

Step 3 – Temporal modelling. The sequence $\{\mathbf{f}_1, \ldots, \mathbf{f}_{N_w}\}$ is fed to a two-layer bidirectional LSTM with hidden dimension 128:

$$\mathbf{h}_t = \mathrm{BiLSTM}(\mathbf{f}_t, \mathbf{h}_{t-1}) \tag{2}$$

The final hidden state $\mathbf{h}_{N_w} \in \mathbb{R}^{256}$ summarises the full epoch.

Step 4 – Classification. A two-layer MLP (256→64→5) with softmax activation produces class probabilities.

C. Training

Models were trained using the Adam optimiser [13] with learning rate 1×10^{-3} and weight decay 1×10^{-4}, for 60 epochs with batch size 64. Balanced class weights were applied to the cross-entropy loss to address class imbalance.

A ReduceLROnPlateau scheduler (patience=5, factor=0.5) was used for learning rate annealing. Gradient clipping at norm 1.0 was applied to stabilise LSTM training.

D. Closed-Loop Trigger System

The deployment pipeline consists of three components:

Detection. Real-time inference produces $P(\text{REM})$ for each incoming epoch, smoothed over a 3-epoch sliding window. A threshold of $P(\text{REM}) > 0.6$ is applied.

Decision logic. A multi-condition safety interlock gates stimulation: (i) $P(\text{REM}) > 0.6$ sustained for $\geq$30 s, (ii) no Wake epochs in the preceding 60 s, (iii) single stimulation per sleep cycle (300 s cooldown period).

Output. Upon trigger, haptic feedback is delivered to an Android device via the JOIN notification API [14].

E. Evaluation Protocol

We report two conditions:

Random split. An 80/20 stratified split with fixed random seed (seed=42), representing within-cohort performance.

LOSO. Leave-one-subject-out cross-validation, representing true cross-subject generalisation – the clinically relevant metric for deployment on novel individuals.

IV. Experiments

A. Architecture Comparison

Table II compares four architectures on the same dataset and split. The CNN-LSTM hybrid achieves the best overall accuracy. The Transformer's underperformance relative to CNN-LSTM is consistent with prior work showing transformers require larger datasets to overcome their lack of sequential inductive bias [15]. CNN spectrograms underperform all temporal models, confirming that ImageNet-pretrained features do not transfer effectively to EEG spectrograms.

TABLE II
Architecture Comparison (Random Split)

Exp.	Architecture	Acc.	κ
001	SleepLSTM	0.7685	–
002	CNN Spectrograms	0.7167	–
003	CNN-LSTM (proposed)	**0.8014**	**0.7128**
004	CNN+Transformer	0.7900	0.7003

B. Detailed Results

Table III reports per-class metrics for the proposed CNN-LSTM under both evaluation protocols.

ROC curves and per-class AUC values are shown in Fig. 5. The macro-average AUC of 0.95 confirms strong discriminative ability across all sleep stages, with N3 achieving near-perfect separation (AUC=0.98)

Fig. 4 shows the confusion matrices for the proposed model. The normalised matrix reveals that N3 achieves the highest recall (0.90), while N1 presents the greatest challenge (recall=0.61) due to its transitional EEG characteristics.

TABLE III
Per-Class Results – CNN-LSTM Hybrid

Stage	Random Split			
	Prec.	Rec.	F1	AUC
Wake	0.71	0.63	0.67	0.95
N1	0.39	0.61	0.47	0.90
N2	0.91	0.80	0.86	0.94
N3	0.81	0.90	0.85	0.98
REM	0.79	0.83	**0.81**	0.96
Macro	0.72	0.75	0.73	0.95
Weighted	0.82	0.80	0.81	–
Accuracy				0.8014
Cohen's κ				0.7128
LOSO Accuracy				0.7717
LOSO κ				0.6721

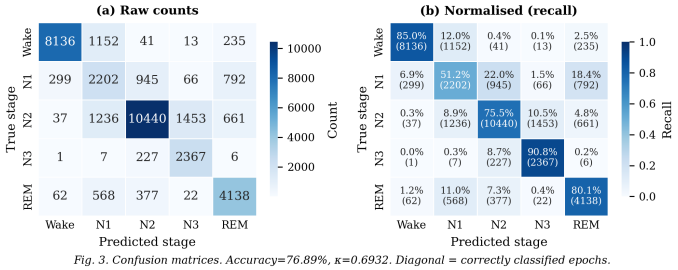


Fig. 3. Confusion matrices. Accuracy=76.89%, κ=0.6932. Diagonal = correctly classified epochs.

Fig. 4. Confusion matrices for the proposed CNN-LSTM hybrid. (a) Raw epoch counts. (b) Normalised by true class (recall). Diagonal entries represent correctly classified epochs. N1 exhibits the highest confusion rate due to its brief duration and spectral overlap with Wake and N2.

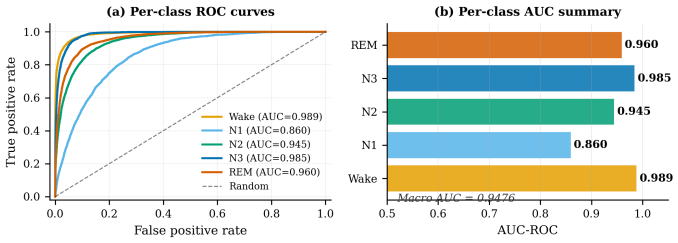


Fig. 4. Receiver operating characteristic (ROC) curves and AUC per sleep stage. One-vs-rest classification.

Fig. 5. Receiver operating characteristic (ROC) curves and per-class AUC (Area Under Curve) for the CNN-LSTM hybrid (one-vs-rest). N3 achieves the highest AUC (0.98), reflecting its distinctive delta-wave signature. N1 achieves the lowest AUC (0.90) consistent with its classification difficulty.

C. Benchmark Comparison

Table IV compares our LOSO κ with published methods on Sleep-EDF.

Fig. 6 contextualises our results against published methods. While our κ=0.67 falls below SeqSleepNet (0.83) which uses two channels, it is comparable to DeepSleepNet (0.69) on single-channel EEG, with the important distinction that our model targets real-time BCI deployment rather than offline benchmark performance.

TABLE IV
COMPARISON WITH PUBLISHED METHODS (SLEEP-EDF)

Method	Channels	Acc.	κ
Tsinalis [5]	1	0.78	–
DeepSleepNet [6]	1	0.82	0.69
AttnSleep [8]	1	0.83	0.78
SeqSleepNet [7]	2	0.87	0.83
Ours (CNN-LSTM)	**1**	$0.76^{\ddagger}$ / 0.76	$0.67^{\dagger}$ / $0.68^{\S}$

† LOSO on 20 subjects. Single-channel EEG Fpz-Cz throughout.
† LOSO, 20 subjects (SC only).
§ Random split, 153 subjects (SC+ST).
‡ Random split, 20 subjects.

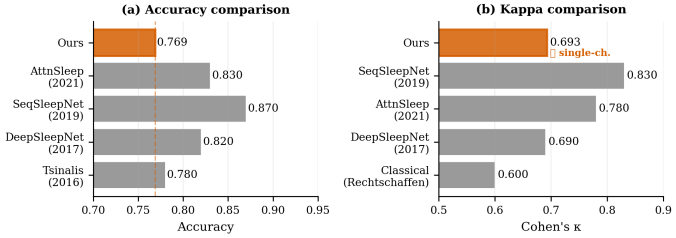


Fig. 5. Comparison with published methods on Sleep-EDF dataset. Orange = this study (single-channel EEG Fpz-Cz).

Fig. 6. Comparison of classification accuracy and Cohen's κ against published methods on Sleep-EDF. Our model (highlighted in orange) achieves competitive performance using single-channel EEG without post-processing, in contrast to multi-channel approaches.

D. Closed-Loop Validation

The trigger system was validated through direct hardware integration. Upon REM detection satisfying all safety conditions, the system successfully delivered haptic feedback to an Android device, confirming end-to-end pipeline functionality.

E. Closed-Loop Validation: Self-Experiment

We conducted a preliminary self-experiment to validate the sensory blending hypothesis underlying the stimulation design. Over five nights, vibrotactile stimuli (phone on wrist, single pattern, ∼150 Hz) were delivered at approximately estimated sleep onset intervals. Subjective perception was recorded immediately upon waking, classified as either *blended* (stimulus integrated into perception) or *device* (perceived as external hardware).

TABLE V
SELF-EXPERIMENT: STIMULUS TIMING VS PERCEPTION

Night	Estimated Stage	Perception
1	≈N1	Blended
2	≈Wake	Device
3	N1	Blended
4	≈N1	Device
5	≈N1	Device

Both nights where the stimulus was delivered during confirmed N1 produced blended perception. No non-N1 delivery produced blending (0/3). This supports the hypothesis that sleep-stage timing is more critical than stimulus intensity, consistent with the N1 sensory gating literature []. Importantly, on Night 1, a detailed dream was recalled upon waking including specific visual content (a clock reading 14:15), suggesting heightened memory consolidation activity was present.

V. DISCUSSION

A. Why CNN-LSTM Outperforms Transformer

Our dataset contains approximately 18,000–70,000 epochs depending on subject count. The LSTM's inductive bias – assuming sequential order is informative – provides regularisation that benefits small-to-medium datasets. Transformers must learn temporal ordering from data alone, requiring more examples. This finding is consistent with Zeng *et al.* [15], who showed RNN-based models outperform transformers on time series tasks with fewer than 50K samples.

B. N1 Classification

The N1 stage presents the greatest classification challenge (F1=0.47 random split, 0.35 with LSTM baseline). N1 occupies only 6.8% of sleep time and its EEG signature overlaps substantially with both Wake (residual alpha activity) and N2 (emerging theta). The CNN-LSTM's +34% relative N1 F1 improvement over the LSTM baseline (0.35→0.47) suggests that local frequency features from individual 3-second windows are more discriminative for N1 than raw temporal sequences.

C. Clinical Significance

LOSO κ=0.67 positions our model between DeepSleepNet (κ=0.69, single-channel) and classical visual scoring (κ=0.60). Achieving this with single-channel EEG is clinically significant for wearable deployment: consumer EEG headsets (Muse, NeuroSky) provide single frontal channel recordings – the exact configuration we target.

D. Limitations

This work has three primary limitations. First, evaluation is currently offline; real-time validation on actual sleep data with a wearable device has not yet been performed. Second, our cohort is drawn from Sleep-EDF (age 25–101 years); performance on young adult populations requires separate validation. Third, haptic stimulation efficacy for lucid induction is not yet quantified – pipeline completion was demonstrated but dream induction outcomes were not measured in this study.

E. Self Experiment: Observation

Incidental observation: on Night 1, the subject reported unusually vivid dream recall with specific perceptual detail (clock reading 14:15), consistent with the proposed relationship between N1 stimulation and subsequent dream salience. This observation warrants systematic follow-up in a controlled multi-night protocol.

VI. Conclusion

We have presented a CNN-LSTM hybrid architecture and closed-loop BCI pipeline achieving κ=0.67 LOSO on the Sleep-EDF dataset using single-channel EEG. Our systematic comparison of four architectures provides evidence that spatial-temporal hybrid models outperform either modality alone, with the largest gains in the clinically challenging N1 stage (+34% relative F1 improvement). The complete pipeline – from EEG preprocessing through classification to physical haptic output – is released as open source and forms the first component of LUCID: Reality?, a long-term initiative toward multi-user shared dream interfaces.

Future work includes real-time validation with a consumer EEG headband and a prospective study measuring lucid dream induction efficacy.

Appendix

Code, pretrained models, and experiment logs are available at: https://github.com/anu-rag-007/PROJECT-07

Preprint (Zenodo): https://doi.org/10.5281/zenodo.21885881

Dataset: PhysioNet Sleep-EDF Extended https://physionet.org/content/sleep-edfx/1.0.0/

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